



Python programming

for biologists

05.09.2026



Our python course team



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Course structure

Meetings

- 13 lectures
- Extra meetings (QA, additional topics)

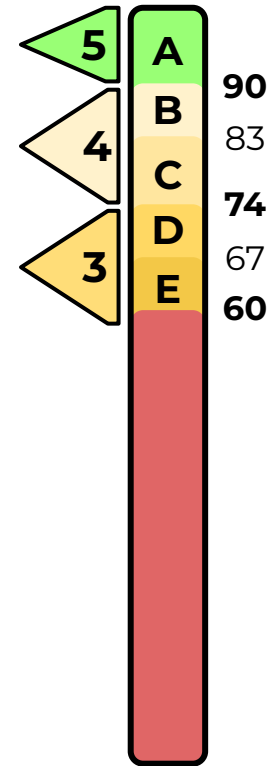
Quizzes

- 3 quizzes (at the beginning of the lectures)
- max 10 points (0: 0%, 5: <50%, 10: >50%)

Homeworks

- Each - max 100 points
- **Deadline: 1 week**
- Submissions after deadline - half of the points
- **Final final deadline: December 5**

Grading



Course program

Basic python 	1. Intro. Git, GitHub 2. Python recap, data types 3. Functions 4. Modules and libraries 5. Files 6. Virtual environments 7. Regular expressions
Scientific data analysis 	8. Numpy 9. Pandas 10. Visualisation 11. Statistics 12. Discussion

next semester...



Advanced python

- OOP, classes
- Decorators
- Iterators & generators
- Web scraping

Tools development

- Parallel programming
- Profiling, performance
- SQL



Sources:

- [Stepik python course](#) (BI)
- [Stepik python course](#) (BEEGEEK)
- [Python tutor](#)
- [LearningGitBranching](#)
- [Hexlet Git course](#)

Soft

- [Python 3.11](#)
- [Jupyter](#)
- [Git](#)
- [PyCharm](#) / [VSCode](#)

Any questions?

Assignments?

Course program?

Grading?

Future plans?





Version control systems

Git

05.09.2026



What do you know about Version Control Systems?

What are the VCS?

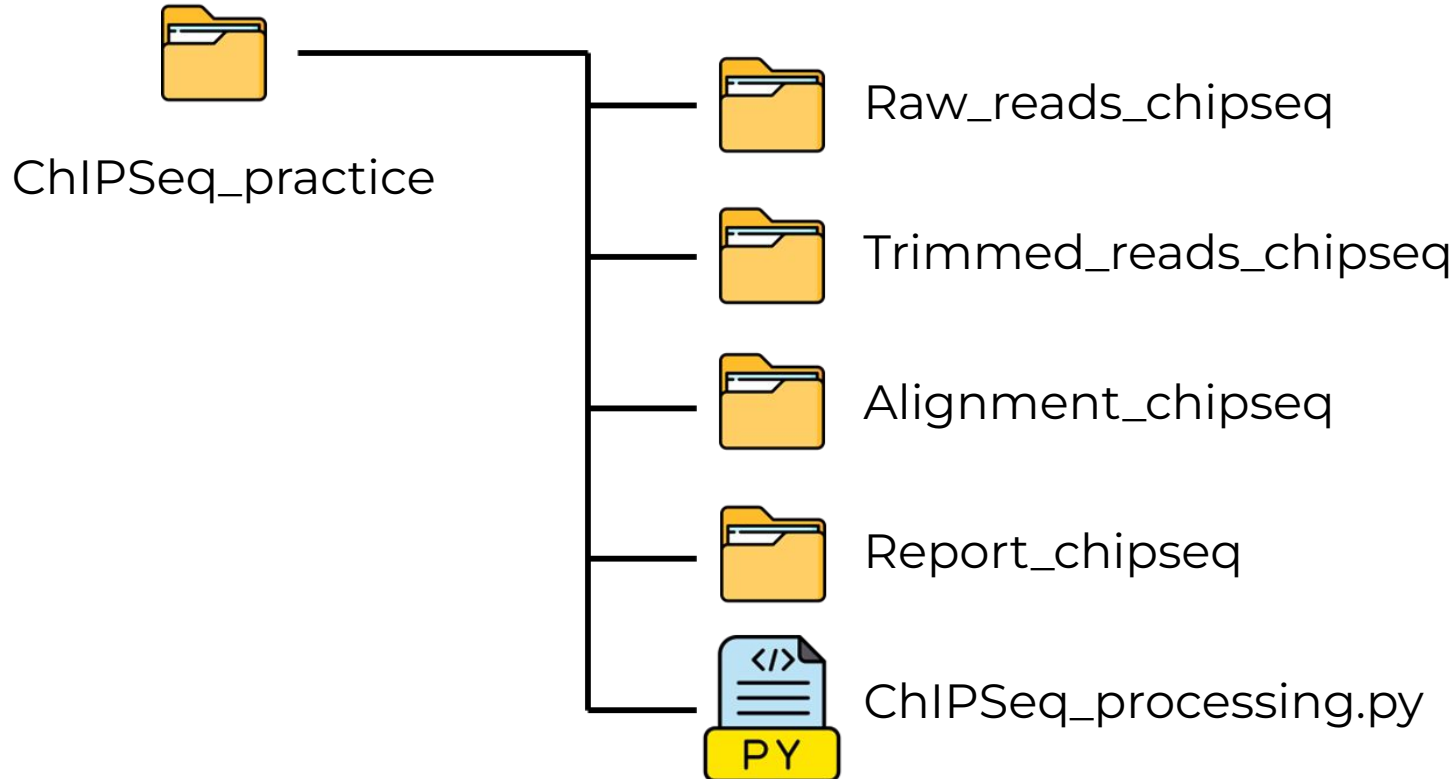
What systems do you know?

Do you use any of them?



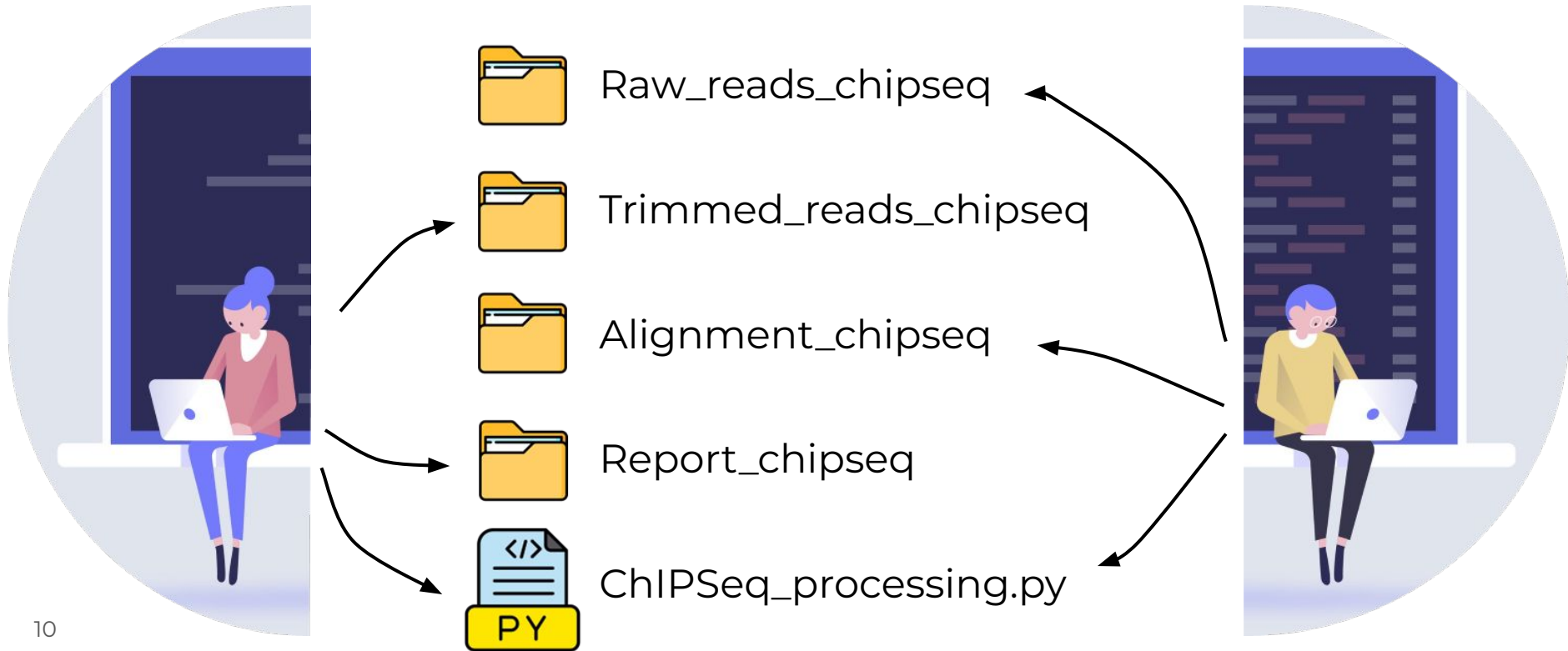


Some bio project





Some bio project



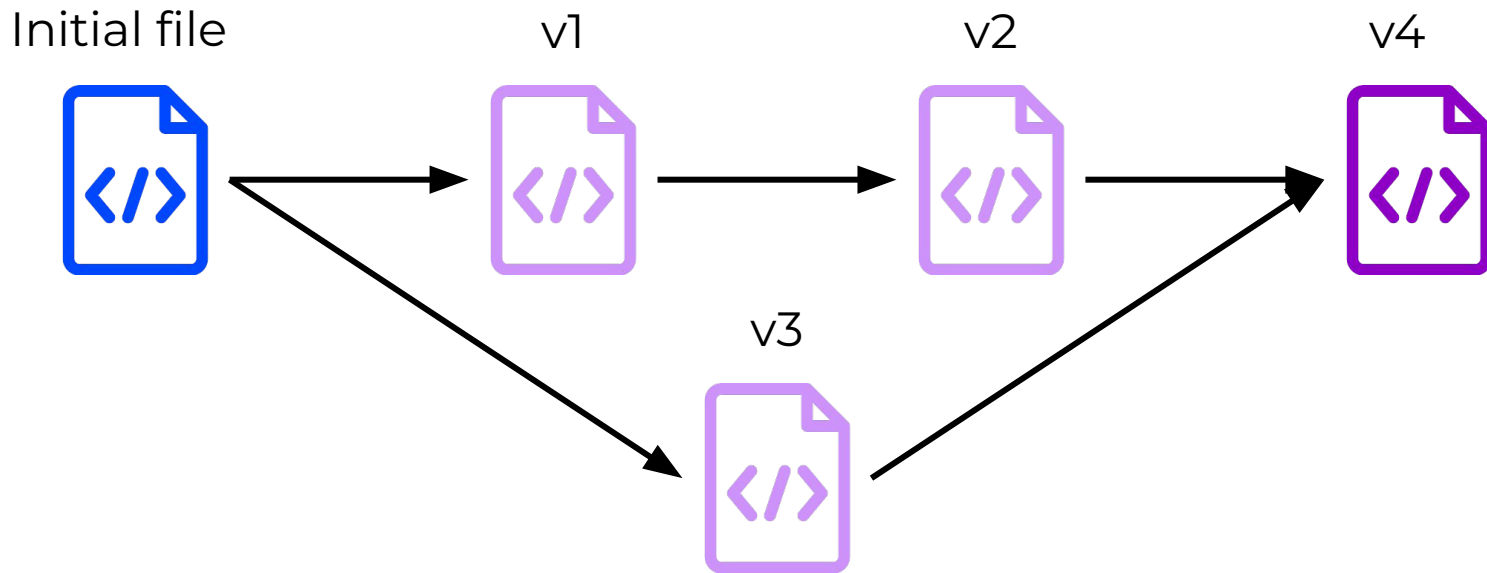


Two main VCS purposes



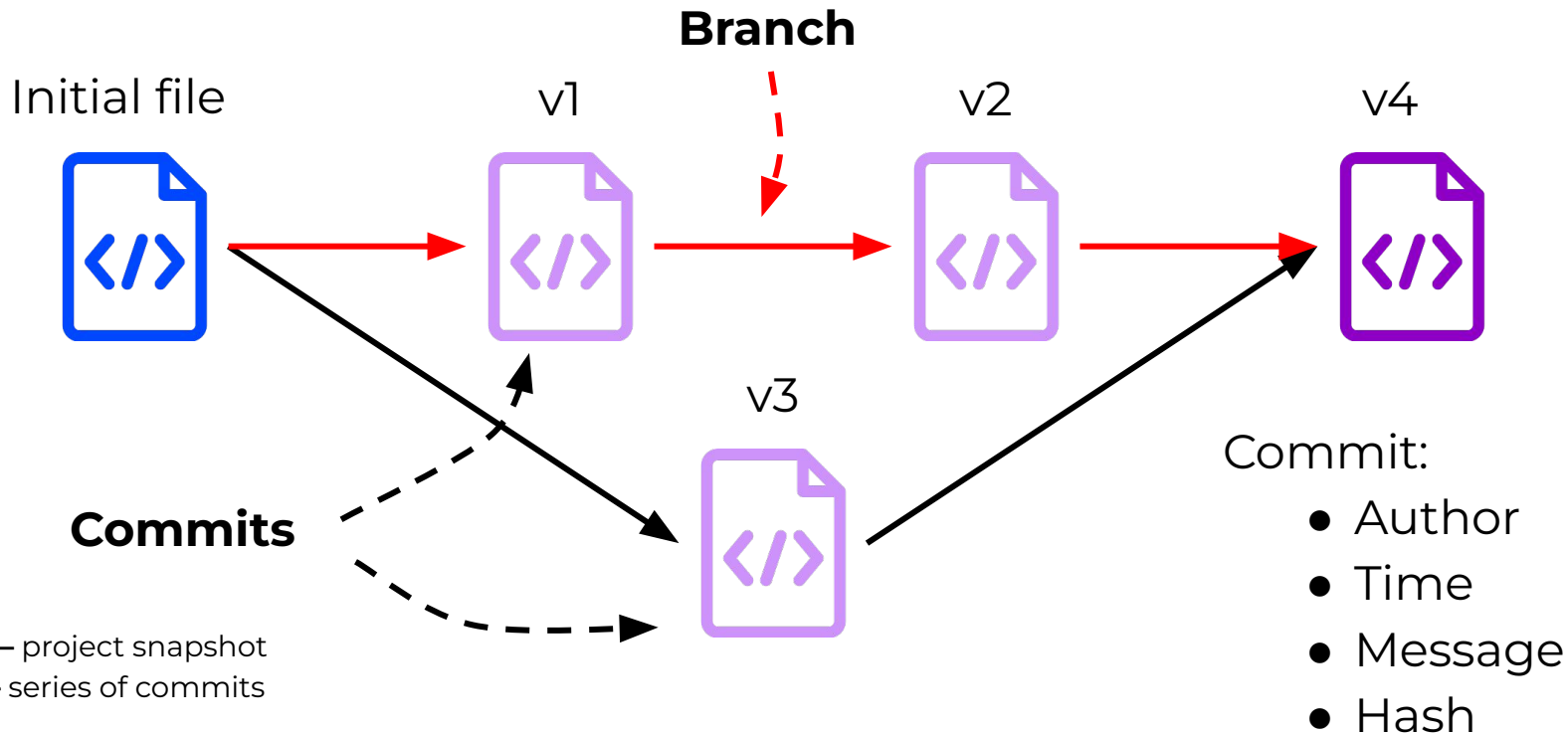


Version controlling





Version controlling

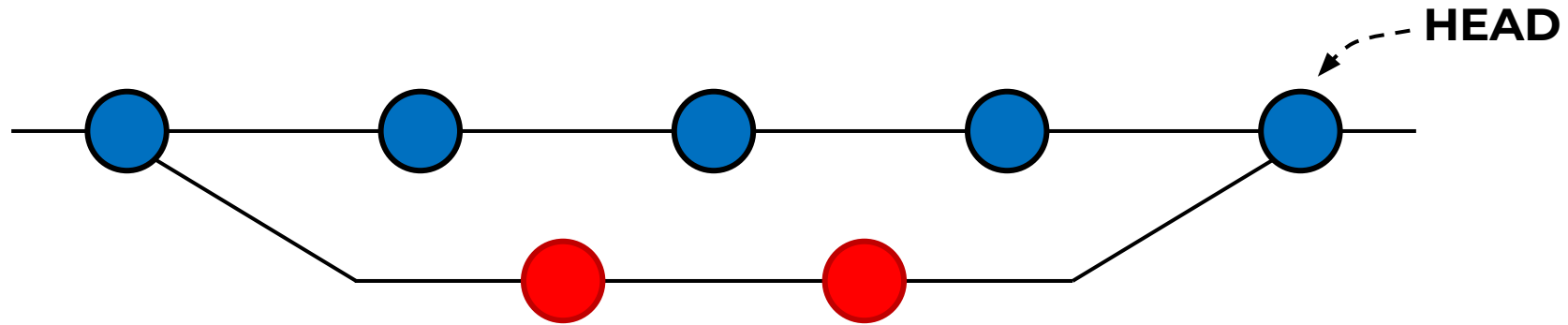


Commit – project snapshot

Branch – series of commits



Version controlling



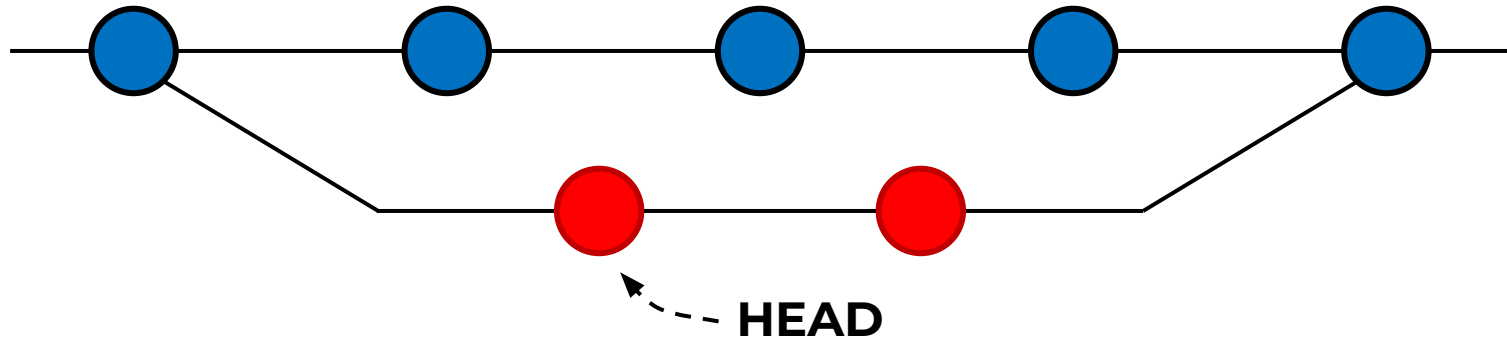
Commit – project snapshot

Branch – series of commits

HEAD – commit you are currently on



Version controlling



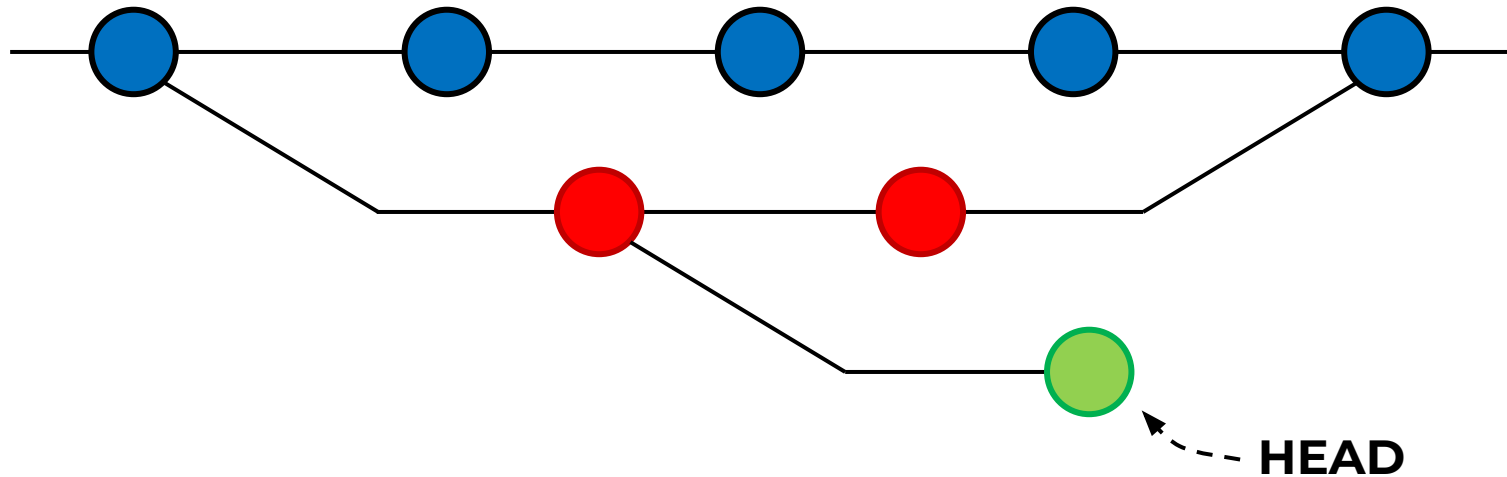
Commit – project snapshot

Branch – series of commits

HEAD – commit you are currently on



Version controlling



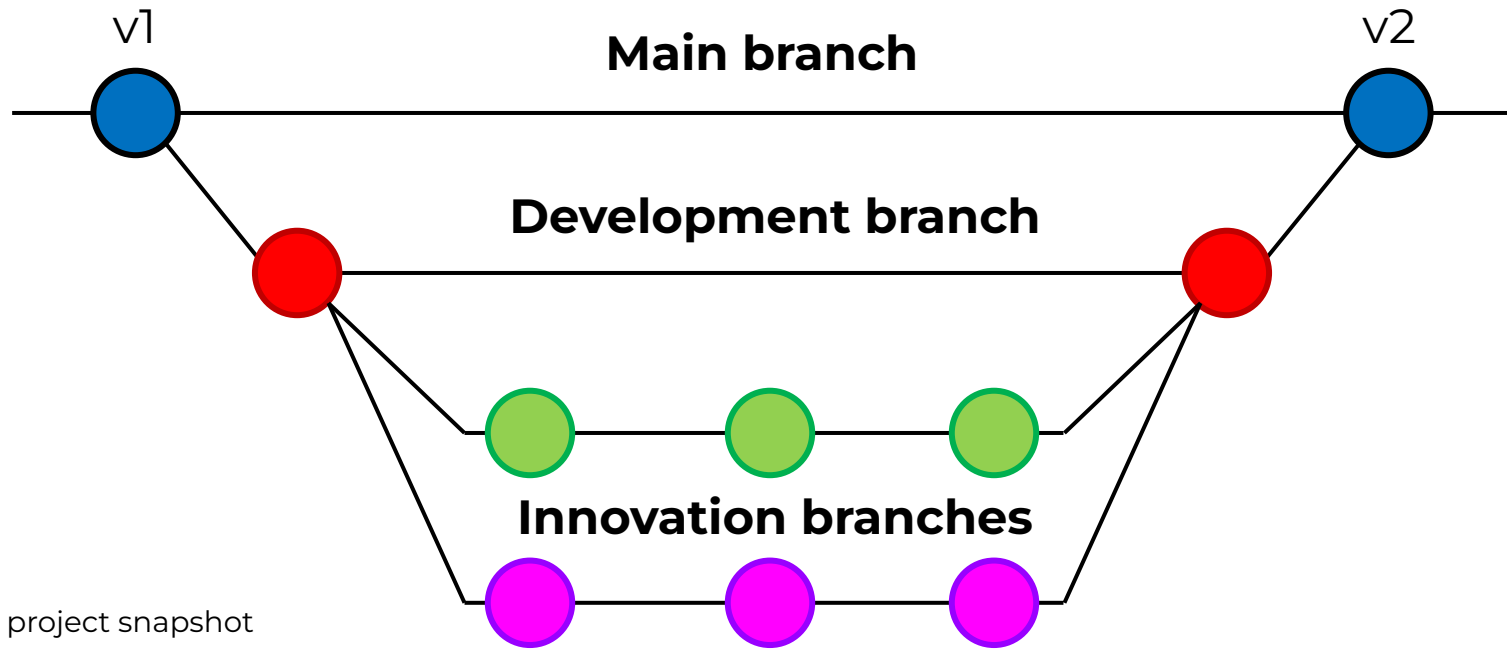
Commit – project snapshot

Branch – series of commits

HEAD – commit you are currently on



Ideal project



Commit – project snapshot

Branch – series of commits

HEAD – commit you are currently on



Git practice

—— Hands on your keyboard ——

Starting new project

```
mkdir test_project
```

Make new empty folder

```
cd test_project
```

Move into it

```
ls -al
```

Check it is empty

```
git init
```

Initialize git **repository**

```
ls -al
```

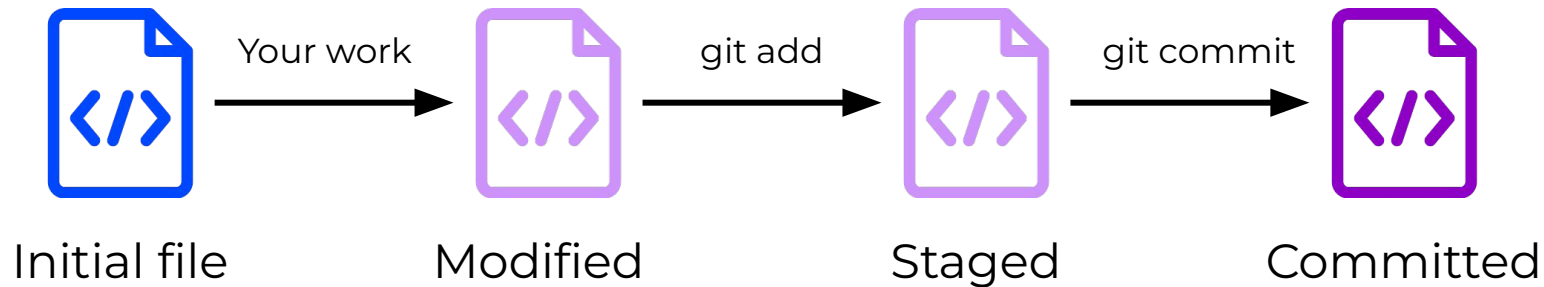
Check once again

What do you see?





Life of the file





Git practice

— Hands on your keyboard —

Working with files

```
echo '...' > file
```

Edit the file

```
git add file
```

Track the file

```
git commit
```

Save changes

```
git config
```

Set up some git things



Commit message rules

- Use **imperative** mood
- Start with **Capital** letter
- Line size < **50** symbols
- Be **informative**
- Do not end with a period
- Use the body if you need

`"Add read_fasta function"`





Git practice

— Hands on your keyboard —

Getting information

`git show`

List of the files

`git status`

List of the files and their statuses

`git log`

Commits history

`git reflog`

List of operations

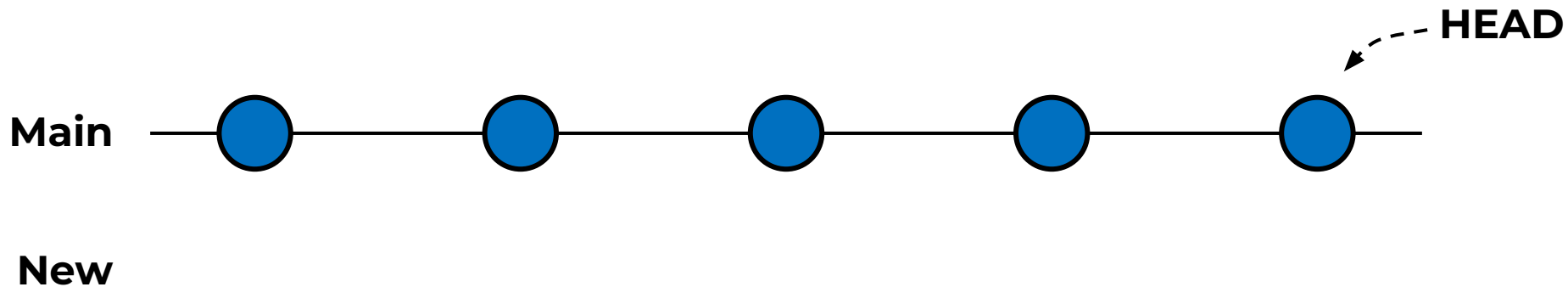
`git blame`

Who edited the file?



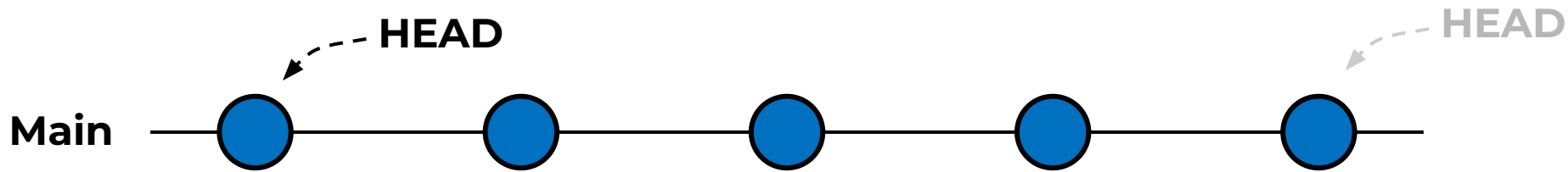


Branching





Branching

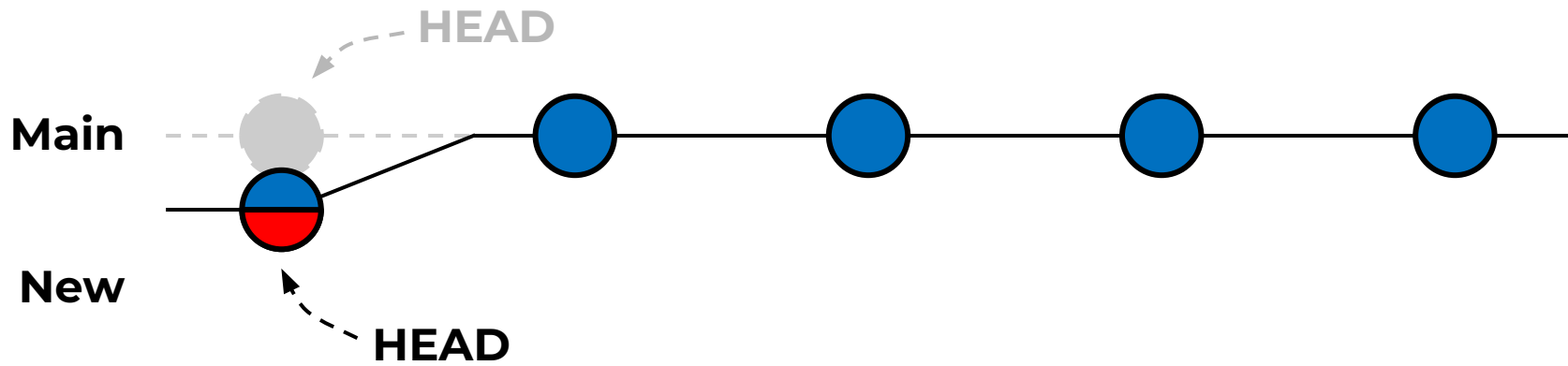


New

```
git checkout <hash>
```



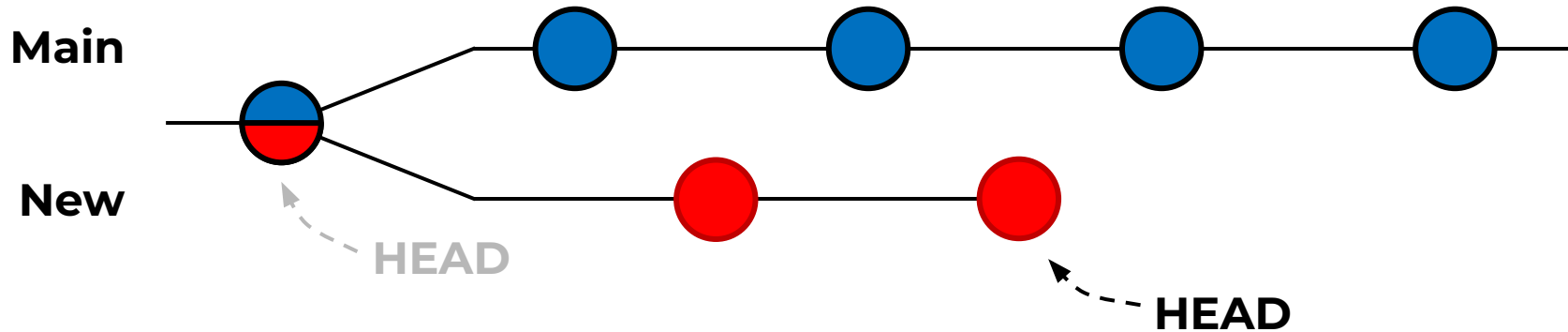
Branching



```
git checkout -b <branch name>
```



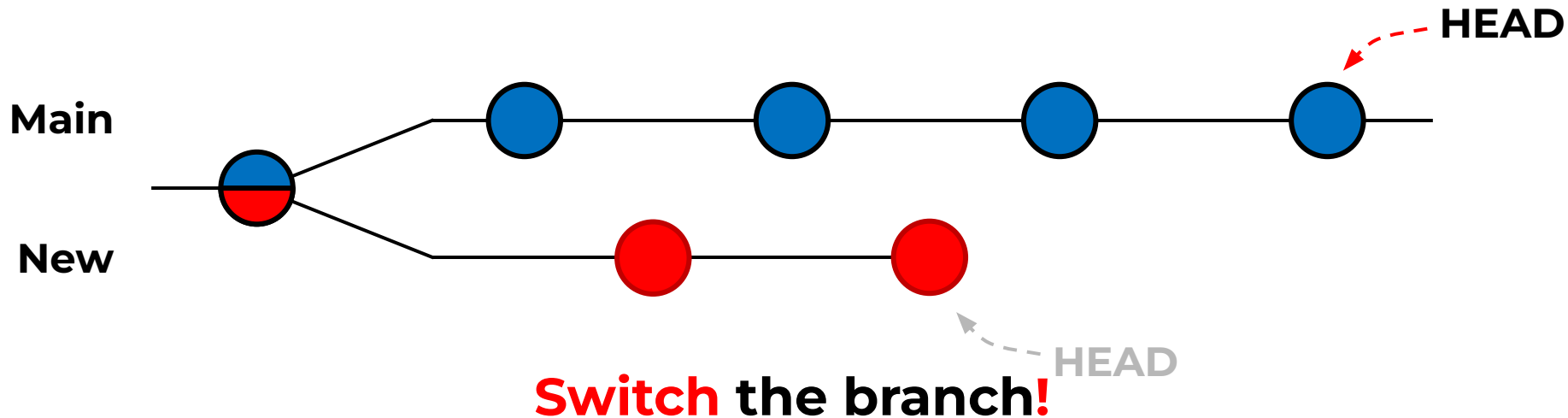
Branching



```
git add && git commit
```



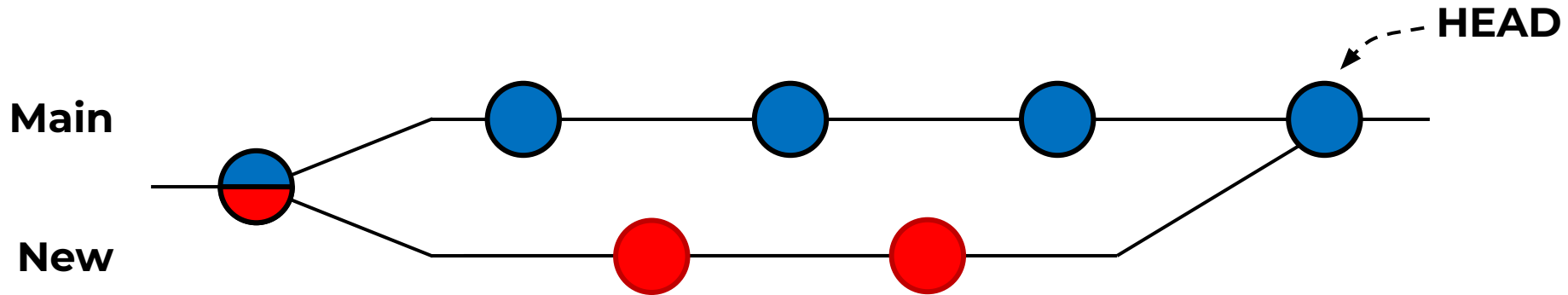
Branching



```
git checkout main
```



Branching



```
git merge <branch name>
```



Git practice

Git branching

`git checkout`

Walk on the commits

`git checkout <hash>`

Switch to the commit

`git checkout <branch>`

Switch to the latest commit

`git branch`

List of the branches

`git branch <name>`

Create the branch

`git checkout -b <name>`

Create the branch and move on

`git merge <name>`

Merges the other branch into yours

`git branch -d <name>`

Delete branch





GitHub

05.09.2026



Remote repositories



You can use them for:

- Code sharing
- Files sharing (even with no coding)
- Resume
- Look for others repo's



GitLab



GitHub



Connecting GitHub with your PC

—— Hands on your keyboard ——

```
ls -al ~/.ssh
```

```
ssh-keygen -t ed25519 -C "your@email.com"
```

```
eval "$(ssh-agent -s)"
```

```
ssh-add ~/.ssh/id_ed25519
```

```
cat ~/.ssh/id_ed25519.pub
```





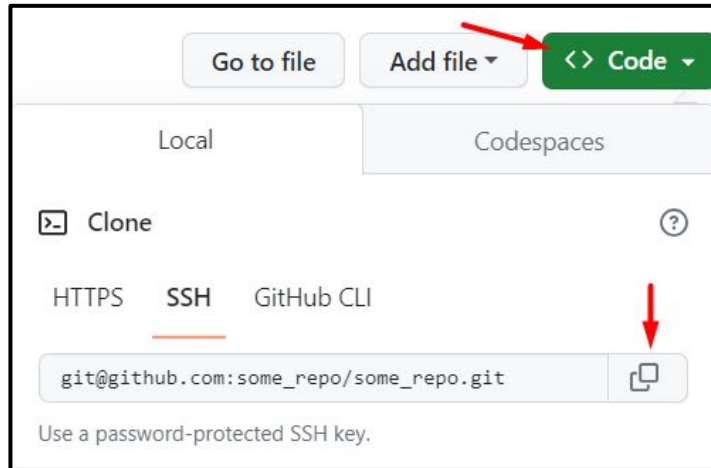
Get remote repo as local

—— Hands on your keyboard ——

```
git clone <SSH_URL>
```

Download the repo

```
cd <directory>
```





Get local repo as remote

—— Hands on your keyboard ——

```
git remote add origin <SSH_URL>
```

```
git remote -v
```

Check URL





Working with remote repo's

`git clone`

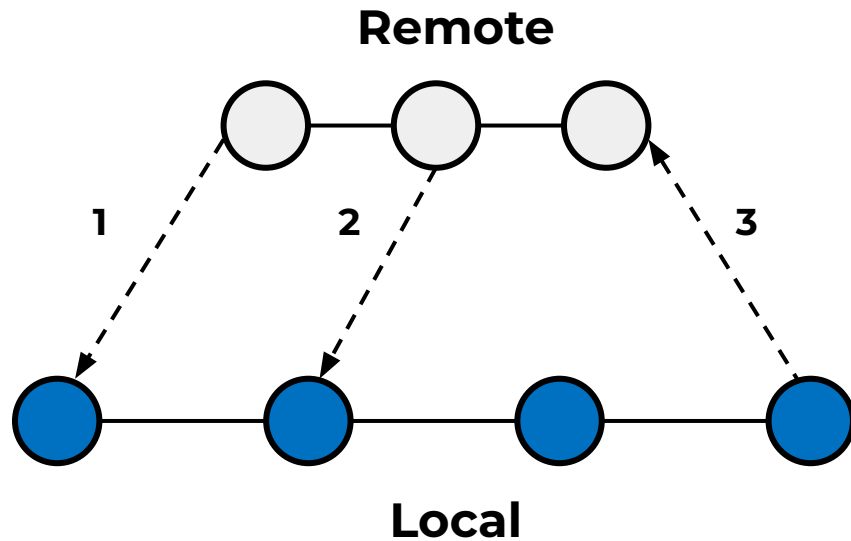
1. **Download** the repo

`git pull`

2. **Update** the repo

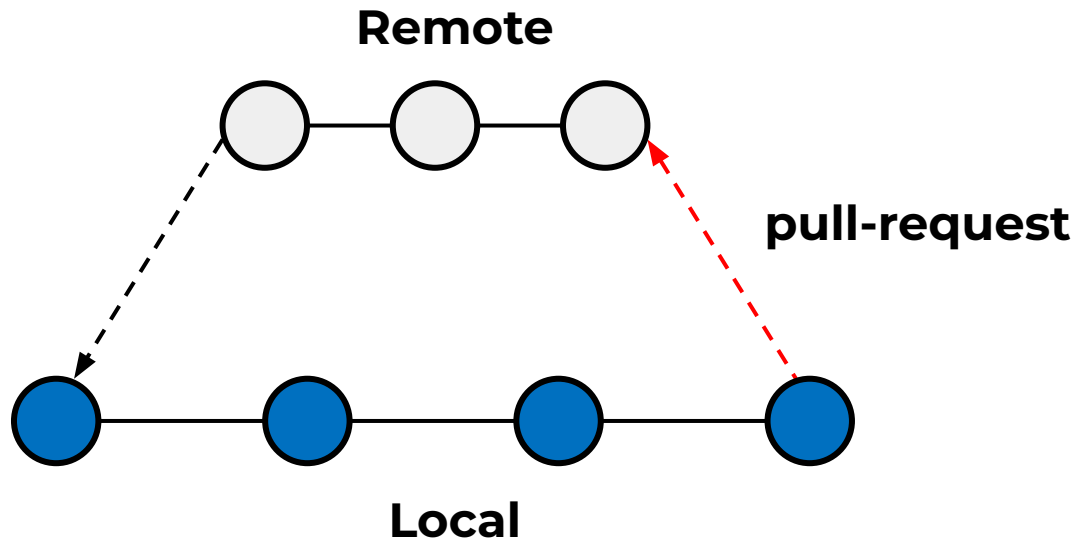
`git push`

3. **Upload** the repo





Merging changes to remote





Troubleshooting

06.09.2025



Git does not commit

Message:

```
# Changes not staged for commit  
(some other text here)
```

Solution:

Pay attention to solution,
suggested by Git.
Only added files can be
committed.



The only correct order:

```
git add
```

```
git commit
```



Over-changing branches

Trouble:

Your branch is ahead of
'origin/master' by 1 commit

Solution:

Commit it with message "WIP"
(work in progress) and do
checkout. In this case you will
not lose something.



```
git commit -m "WIP"
```




My branch is ahead of origin

Message:

"I have something modified and not ready for commit, but I need to change branch"

Solution:

It's not an error ;)
Git noticed that you have 1 commit that it does not see on remote repository and suggest you to **push** the changes.



```
git push origin <branch>
```



My HEAD is detached

Message:

You are in 'detached HEAD' state.

Solution:

Probably, you face with it while review your **old commit**. You can go back to the old commit (discard changes) or save them as new branch



[Stackoverflow](https://stackoverflow.com)



Undo, undo, undo!

2.4 Git Basics - Undoing Things

Trouble:

"Last commit is wrong and I need to remove it".

Solutions:

```
git commit --amend
```

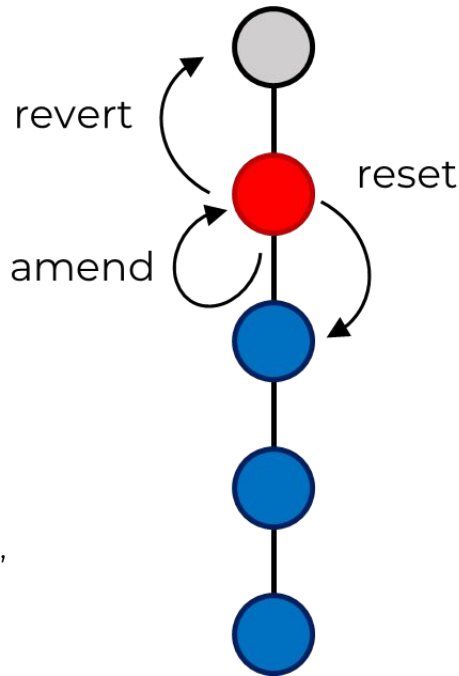
Add more files to commit

```
git reset HEAD~1
```

Move HEAD 1 commit back,
has **soft** and **hard** modes

```
git revert
```

Create new one, "anti-commit"



Our successes

- We now know what is Git and GitHub
- We can manage Git projects
- We can link local and remote repo's
- We can себя показать и других посмотреть on GitHub



Any questions?

Course structure?

Git usage?

GitHub?

